

Eugene Katsevich

Assistant Professor of Statistics and Data Science

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Employment

University of Pennsylvania, Department of Statistics and Data Science, Assistant Professor, 2020-present.

- Member of the graduate group in Applied Mathematics and Computational Science (AMCS)
- Member of the graduate group in Genomics and Computational Biology (GCB)

Carnegie Mellon University, Department of Statistics and Data Science, Postdoctoral Researcher, 2019-2020.

Advisors: Kathryn Roeder and Aaditya Ramdas.

Education

Stanford University, Ph.D. in Statistics, 2019.

Thesis Advisor: Chiara Sabatti.

Princeton University, A.B. in Mathematics (with Highest Honors), 2014.

Thesis Advisor: Amit Singer.

Publications

*Equal contribution.

Working manuscripts

28. Z. Niu, Y. He, J. Galante, A. R. Gschwind, J. Ray, L. M. Steinmetz, J. M. Engreitz, and **E. Katsevich**. PerturbPlan: An analytical framework for designing Perturb-seq experiments. Available on [bioRxiv](#).
27. L. Deutsch, T. Barry, **E. Katsevich**. Scaling sceptre to large perturb-seq datasets. In preparation.
26. T. Barry, Z. Niu, **E. Katsevich**, X. Lin. The permuted score test for robust differential expression analysis. Available on [arXiv](#).
25. A. Chakraborty, J. Zhang, and **E. Katsevich**. Doubly robust and computationally efficient high-dimensional variable selection. Available on [arXiv](#).

Manuscripts under review

24. A. Chakraborty*, J. Lee*, **E. Katsevich**. Power of masking methods for adaptive testing in a multivariate normal means problem. Available on [arXiv](#). Under review at *Biometrika*.
23. J. Ray, E. Jagoda, M. U. Sheth, J. Galante, D. Amgalan, E. Mattei, A. R. Gschwind, C. J. Munger, J. Huang, G. Munson, M. Murphy, T. Barry, V. Singh, A. Baskaran, H. Kang, R. Issner, C. B. Epstein, F. J. Najm, **E. Katsevich**, E. Gaskell, L. M. Steinmetz, and J. M. Engreitz. A systematic survey of distal element-gene regulatory interactions with Direct-Capture Targeted Perturb-seq. Under review at *Nature Genetics*. Available on [bioRxiv](#).
22. P. J. Law, J. Vijaykrishnan, J. Smith, T. Barry, M. Mandelia, **E. Katsevich**, and R. S. Houlston. Enhancer-gene regulatory interactions in colorectal cancer revealed through genome-wide CRISPRi perturbations. *Wellcome Open Research*, 2026 (open peer review). Available at [Wellcome Open Research](#).

Published or forthcoming papers

21. Z. Niu, Z. Huang, J. Ray Choudhury, and **E. Katsevich**. Saddlepoint approximations for plug-in resampling. Available on [arXiv](#). *Biometrika*, forthcoming.
20. L. Deutsch and **E. Katsevich**. Location tests with noisy proxies for latent variables. *Statistics and Probability Letters*, 2025. Available on [Science Direct](#).
19. M. Conery, J. A. Pippin, Y. Wagley, K. Trang, M. C. Pahl, D. A. Villani, L. J. Favazzo, C. L. Ackert-Bicknell, M. J. Zuscik, **E. Katsevich**, A. D. Wells, B. S. Zemel, B. F. Voight, K. D. Hankenson, A. Chesi, and S. F. A. Grant. GWAS-informed data integration and non-coding CRISPRi screen illuminate genetic etiology of bone mineral density. *Genome Biology*, 2025. Available at [Genome Biology](#).
18. R. E. Yan, A. Corman, L. Katgara, X. Wang, X. Xue, Z. Z. Gajic, R. Sam, M. Farid, S. M. Friedman, J. Choo, I. Raimondi, S. Ganesan, **E. Katsevich**, J. P. Greenfield, N. Dahmane, and N. E. Sanjana. Pooled CRISPR screens with joint single-nucleus chromatin accessibility and transcriptome profiling. *Nature Biotechnology*, 2025. Available at [Nature Biotechnology](#).
17. T. Barry, K. Mason, K. Roeder, and **E. Katsevich**. Robust differential expression testing for single-cell CRISPR screens at low multiplicity of infection. *Genome Biology*, 2024. Available at [Genome Biology](#).
16. T. Barry, K. Roeder, and **E. Katsevich**. Exponential family measurement error models for single-cell CRISPR screens. *Biostatistics*, 2024. Available at [Oxford University Press](#).
15. Z. Niu, A. Chakraborty, O. Dukes, and **E. Katsevich**. Reconciling model-X and doubly robust approaches to conditional independence testing. *Annals of Statistics*, 52(3):895–921, 2024. Available at [Project Euclid](#).
14. J. A. Morris, C. Caragine, Z. Daniloski, J. Domingo, T. Barry, L. Lu, K. Davis, M. Ziosi, D. A. Glinos, S. Hao, E. P. Mimitou, P. Smibert, K. Roeder, **E. Katsevich**, T. Lappalainen, and N. E. Sanjana. Discovery of target genes and pathways at GWAS loci by pooled single-cell CRISPR screens. *Science*, 2023. Available at [Science](#).
13. J. Tian, X. Chen, **E. Katsevich**, J. Goeman, and A. Ramdas. Large-scale simultaneous inference under dependence. *Scandinavian Journal of Statistics*, 50(2): 750–796, 2023. Available at [Wiley](#).
12. **E. Katsevich**, C. Sabatti, and M. Bogomolov. Filtering the rejection set while preserving false discovery rate control. *Journal of the American Statistical Association*, 118(541): 165–176, 2023. Available at [Taylor and Francis Online](#).

11. **E. Katsevich** and A. Ramdas. On the power of conditional independence testing under model-X. *Electronic Journal of Statistics*, 16(2): 6348–6394, 2022. Available on [Project Euclid](#).
10. M. Liu, **E. Katsevich**, L. Janson, and A. Ramdas. Fast and powerful conditional randomization testing via distillation. *Biometrika*, 109(2):277–293, 2022. Available at [Oxford University Press](#).
9. T. Barry, X. Wang, J. A. Morris, K. Roeder, and **E. Katsevich**. SCEPTRE improves calibration and sensitivity in single-cell CRISPR screen analysis. *Genome Biology*, 22:344, 2021. Available at [Genome Biology](#).
8. **E. Katsevich** and A. Ramdas. Simultaneous high-probability bounds on the false discovery proportion in structured, regression, and online settings. *Annals of Statistics*, 48(6):3465–3487, 2020. Available at [Project Euclid](#).
7. M. Sesia, **E. Katsevich**, S. Bates, E. Candès, C. Sabatti. Multi-resolution localization of causal variants across the genome. *Nature Communications*, 11:1093, 2020. Available at [nature.com](#).
6. J. Zhu, Q. Zhao, **E. Katsevich**, C. Sabatti. Exploratory Gene Ontology Analysis with Interactive Visualization. *Nature Scientific Reports*, 9:7793, 2019. Available at [nature.com](#).
5. **E. Katsevich** and C. Sabatti. Multilayer Knockoff Filter: Controlled variable selection at multiple resolutions. *Annals of Applied Statistics*, 13(1):1–33, 2019. Available at [Project Euclid](#).
4. J. Anden, **E. Katsevich**, and A. Singer. Covariance estimation using conjugate gradient for 3D classification in cryo-EM. In *IEEE Int Symp Biomed Imaging*, New York, New York, April 2015. Available at [PubMed](#).
3. **E. Katsevich**, A. Katsevich, A. Singer. Covariance matrix estimation for the cryo-EM heterogeneity problem. *SIAM Journal on Imaging Sciences*, 8(1):126–185, 2015. Available at [PubMed](#).
2. B. Shi, **E. Katsevich**, B. Chiang, A. Katsevich, and A. Zamyatin. Image registration for motion estimation in cardiac CT. In *SPIE Medical Imaging*, San Diego, California, February 2014. Available at [SPIE digital library](#).
1. **E. Katsevich**, A. Katsevich, and G. Wang. Stability of the interior problem for polynomial region of interest. *Inverse Problems*, 28(6), 2012. Available at [PubMed](#).

Software

- **sceptre**: R package for rigorously identifying regulatory relationships based on single-cell CRISPR screen experiments ([webpage](#), [manual](#)). This package is the first to be [recommended by 10x Genomics](#) for single-cell CRISPR screen analysis.
- **PerturbPlan**: Shiny app and companion R package for Perturb-seq and TAP-seq experimental design. It helps balance power and cost, with an emphasis on speed, flexibility, and interactivity ([web app](#), [R package](#)).
- **ondisc**: R that facilitates analysis of large-scale single-cell data out-of-core ([webpage](#)).
- **simulatr**: R package and Nextflow pipeline enabling portable and scalable numerical simulations ([webpage](#)).

Recognition

- Claude Merion Faculty Scholar (2025-2026)
- Wharton Teaching Excellence award (2021)
- Jerome H. Friedman Applied Statistics Dissertation Award (2019).
- Statistics Department Teaching Assistant Award (2016).
- Hertz Fellowship (2014-2019).
- National Defense Science and Engineering Fellowship, declined (2014-2017).
- George B. Covington Thesis Prize in Mathematics (2014).
- Early election to Phi Beta Kappa (2013).
- Barry Goldwater Scholarship (2012).
- Shapiro Prize for Academic Excellence (2011, 2012).
- Freshman First Honor Prize (2011).

Awards received by trainees

- Ziang Niu received the Lawrence D. Brown best paper award from the Wharton Statistics and Data Science department for “Reconciling model-X and doubly robust approaches to conditional independence testing.” (2024)
- Timothy Barry received Mihaela Serban Memorial Award for best poster from the Pittsburgh chapter of the American Statistical Association for “Robust differential expression testing for single-cell CRISPR screens at low multiplicity of infection.” (2023)
- Timothy Barry received ASHG Reviewers’ Choice Award for abstract on “SCEPTRE improves calibration and sensitivity in single-cell CRISPR screen analysis.” (2021)

Teaching

Courses Taught at University of Pennsylvania

- STAT 9610: Statistical Methodology (PhD level), Fall ’21, Fall ’22, Fall ’23, Fall ’24.
- STAT 4710: Modern Data Mining (Undergraduate level), Spring ’21, Fall ’21, Fall ’22, Fall ’23.

Courses Taught at Stanford

- STATS 302: Qualifying Exams Workshop (PhD Level), Summer ’17.

Training

Doctoral Student Supervision

- Ziang Niu; jointly with Bhaswar Bhattacharya (Wharton Statistics and Data Science; Fall 2023 – present)
- Louis Deutsch (Wharton Statistics and Data Science; Summer 2023 – present)
- Timothy Barry; jointly with Kathryn Roeder (Carnegie Mellon University; Summer 2020 – Fall 2023)

Dissertation Committee Membership

- Kaishu Mason (Wharton Statistics and Data Science; Spring 2024 – present)
- Abhinav Chakraborty (Wharton Statistics and Data Science; Spring 2024 – 2025)
- Jeffrey Zhang (Wharton Statistics and Data Science; Spring 2024 – 2025)
- Ganesh Chandrasekaran (UPenn Biostatistics; Fall 2023 – present)
- Shannon Laub (UPenn Bioengineering; Fall 2023 – present)

Master's Student Supervision

- Taylor Brysgel (Pharmacology Graduate Group; 2026)
- Changhao Ge (Applied Math and Computational Science; 2026)

Doctoral Student Research Mentorship (Informal)

- Jyotishka Ray Choudhury (Georgia Institute of Technology; Fall 2024 – present)
- Sophia Lu (Stanford University Statistics; Summer 2021)
- Abhinav Chakraborty (Wharton Statistics and Data Science; Fall 2021 – 2025)
- Jeffrey Zhang (Wharton Statistics and Data Science; Spring 2023 – 2025)
- Kaishu Mason (Wharton Statistics and Data Science; Spring 2023)
- Junu Lee (Wharton Statistics and Data Science; Fall 2024 – present)
- Yihui He (Wharton Statistics and Data Science; Fall 2024 – present)
- Zhihan Huang (Wharton Statistics and Data Science; Spring 2026 – present)

Master's Student Research Mentorship (Informal)

- Jyotishka Ray Choudhury (Indian Statistical Institute; Summer 2023 – 2024)
- Ziang Niu (University of Pennsylvania AMCS; Spring 2022 – Summer 2023)

Undergraduate Student Research Mentorship (Informal)

- Winnie Dong (University of Pennsylvania; Spring 2024)

Professional Service Activities

- *Member*, NSF Review Panel (2024)
- *Organizer*, topic-contributed session on “Statistical analysis for pooled single-cell CRISPR screen data” at STATGEN conference in Pittsburgh (2024)
- *Member*, Statistics departmental postdoc selection committee (2021-22, 2022-23)
- *Member*, Statistics PhD admissions committee (2021-22, 2022-23, 2023-24, 2025-26)
- *Organizer*, Wharton statistics departmental seminar (Spring 2022, Spring 2024)

- *Reviewer* (43 manuscripts): *Annals of Statistics* (7), *Journal of the Royal Statistical Society (Series B)* (5), *Journal of the American Statistical Association* (5), *Biometrika* (3), *Bernoulli* (1), *Annals of Applied Statistics* (1), *Statistical Science* (1), *Proceedings of the National Academy of Sciences* (2), *Biometrics* (3), *Statistics and Probability Letters* (1), *Statistics and Computing* (1), *Journal of Computational and Graphical Statistics* (2), *Annals of the Institute of Statistical Mathematics* (1), *Journal of Multivariate Analysis* (1), *Electronic Journal of Statistics* (1), *Science* (1), *Nature Methods* (2), *Cell Systems* (1), *American Journal of Human Genetics* (1), *Genetics* (1), *Genome Medicine* (1), *BMC Bioinformatics* (1).
- *Organizer*, Stanford statistics department orientation program for PhD students (2018).
- *Organizer*, Hertz West Coast Retreat (2017).
- *Academic chair*, Princeton Math Club (2012).
- *Head problem writer*, Princeton University Math Competition (2011).

Presentations

Invited talks

- *PerturbPlan: A web application for perturb-seq experimental design.*
Symposium: AI Meets Statistics: Biomedical Data Perspectives at IVADO, August 20-21, 2026.
- *Fast and accurate perturbation-gene association testing in perturb-seq via the saddlepoint approximation.*
Johns Hopkins Biostatistics Seminar, April 27, 2026.
- *Power of masking methods for adaptive testing in a multivariate normal means problem.*
Harvard University Statistics Seminar, Feb. 23, 2026, held virtually.
- *spaCRT: a resampling-free conditional randomization test.*
International Conference on Statistics and Data Science in Nice, France, Dec. 16-19, 2024.
- *Scaling SCEPTRE to large single-cell CRISPR screens.*
STATGEN 2024 in Pittsburgh, Pennsylvania, May 1-3, 2024.
- *Reconciling model-X and doubly robust approaches to conditional independence testing.*
International Conference on Statistics and Data Science in Lisbon, Portugal, Dec. 18-21, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*
EMBL-EBI industry workshop: Perturb-Seq data analysis and integration in Cambridge, MA, Jun. 21-22, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*
New England Statistics Symposium in Boston, MA, Jun. 6, 2023.
- *Robust differential expression testing for single-cell CRISPR screens.*
Impact of Genomic Variation on Function (IGVF) Consortium CRISPR focus group, May 19, 2023, held virtually.
- *Model-X versus doubly robust conditional independence testing.*
Statistics Seminar at Stanford University, May 2, 2023.
- *Model-X versus doubly robust conditional independence testing.*
Causal Inference Reading Group at Cambridge University, Apr. 28, 2023, held virtually.
- *Reconciling model-X and doubly robust approaches to conditional independence testing.*
International Seminar on Selective Inference, Mar. 1, 2023, held virtually.

- *Statistical Analysis of Single Cell CRISPR Screens.*
Joint Statistical Meetings in Washington, D.C., Aug. 8, 2022.
- *Statistical Analysis of Single Cell CRISPR Screens.*
Statistics seminar at the Data Science and Operations department at the University of Southern California, Apr. 22, 2022, held virtually.
- *Statistical Analysis of Single Cell CRISPR Screens.*
Biostatistics seminar at Yale University,
Feb. 2, 2021, held virtually.
- *Statistical Analysis of Single Cell CRISPR Screens.*
Open Insights in Biomedical Data Science Seminar at the University of Pennsylvania,
Nov. 13, 2020, held virtually.
- *Finite-sample optimality and large-sample power analysis of the conditional randomization test.*
Young Data Science Researcher Seminar at ETH Zurich, Nov. 6, 2020, held virtually.
- *Statistical analysis of single cell CRISPR screens.*
Stanford Biostatistics Workshop, Oct. 8, 2020, held virtually.
- *The conditional randomization test in theory and in practice.*
International Seminar on Selective Inference, Aug. 20, 2020, held virtually.
- *A theoretical treatment of conditional independence testing under Model-X.*
Joint Statistical Meetings, Aug. 2–6, 2020, held virtually.
- *Multiple testing for modern data: structure, curation, and replicability.*
U.C. San Diego Halicioglu Data Science Institute Special Seminar, Feb. 12, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*
Rutgers University Statistics Seminar, Feb. 5, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*
Wharton Statistics Seminar, University of Pennsylvania, Jan. 30, 2019.
- *Multiple testing for modern data: structure, curation, and replicability.*
University of Chicago Statistics Colloquium, Jan. 24, 2019.
- *Controlling FDR while highlighting distinct discoveries, with applications to GO enrichment analysis.*
Stanford University Biostatistics Workshop, Oct. 11, 2018.
- *Controlling FDR while highlighting distinct discoveries, with applications to GO enrichment analysis.*
U.C. Berkeley Statistics and Genomics Seminar, Sep. 27, 2018.

Invited discussions

- *"False discovery proportion envelopes with m -consistency"* by Iqraa Meah.
International Seminar on Selective Inference, Nov. 25, 2024, held virtually.
- *"Simultaneous FDP bounds via knockoffs and closed testing"* by Jinzhou Li.
International Seminar on Selective Inference, Jan. 25, 2023, held virtually.
- *"Detecting Multiple Replicating Signals using Adaptive Filtering Procedures"* by Jingshu Wang.
International Seminar on Selective Inference, May 28, 2020, held virtually.

Contributed talks

- *Power of masking methods for adaptive testing in a multivariate normal means problem.*
Joint Statistical Meetings, Aug. 1–6, 2026, in Boston, Massachusetts.
- *Comparing learn-then-test strategies for multiple testing.*
International Conference on Statistics and Data Science in Seville, Spain, Dec. 15–18, 2025.
- *PerturbPlan: A web application for Perturb-seq experimental design.*
Impact of Genomic Variation on Function Consortium Annual Meeting, Oct. 21–23, 2025 in Durham, North Carolina. One of 12 abstracts selected for podium presentation to entire consortium.
- *spaCRT: a resampling-free conditional randomization test.*
Bernoulli-IMS World Congress in Probability and Statistics, Aug. 12–16, 2024 in Bochum, Germany.
- *A theoretical treatment of conditional independence testing under Model-X.*
Bernoulli-IMS One World Symposium, Aug. 24–28, 2020, held virtually.
- *Simultaneous High-Probability Bounds on the False Discovery Proportion for Nested Sequences of Rejection Sets.*
International Conference on Multiple Comparison Procedures, Dec. 13–15, 2019, in Taipei, Taiwan.
- *Gene Ontology enrichment testing: Reconciling FDR control with filtering.*
Joint Statistical Meetings, Jul. 28–Aug. 2, 2018, in Vancouver, Canada.
- *The multilayer knockoff filter: Controlled multi-resolution variable selection.*
International Conference on Multiple Comparison Procedures, Jun. 20–23, 2017, in Riverside, California.

Poster Presentations

- *PerturbPlan: A web application for perturb-seq experimental design.*
STATGEN 2026: Conference on Statistics in Genomics and Genetics, May 18–20, 2026, in Atlanta, Georgia. Invited poster.
- *Perturbation-expression association analysis in low-MOI single-cell CRISPR screens with SCEPTRE.*
Biology of Genomes, May 9 - 13, 2023, in Cold Spring Harbor Laboratory, New York.
- *SCEPTRE improves calibration in single-cell CRISPR screen analysis.*
Biology of Genomes, May 10 - 14, 2022, in Cold Spring Harbor Laboratory, New York.
- *Conditional resampling improves sensitivity and specificity of genome-wide CRISPR regulatory screens.*
American Society for Human Genetics, Oct. 27–30, 2020, held virtually.
- *Controlling FDR While Highlighting Selected Discoveries.*
International Conference on Multiple Comparison Procedures, Dec. 13–15, 2019, in Taipei, Taiwan.
- *Multi-resolution association analysis for exome-wide sequencing.*
American Society for Human Genetics, Oct. 16–20, 2018, in San Diego, California.
- *Controlling FDR while highlighting distinct discoveries.*
Workshop on Higher-Order Asymptotics and Post-Selection Inference, Sep. 8–10 2018, in St. Louis, Missouri.
- *Multilayer FDR control for genetic association studies.*
Graybill Conference on Statistical Genomics and Genetics, Jun. 5–7 2017, in Fort Collins, Colorado. Best student poster award.
- *The multilayer knockoff filter: Multilayer FDR control for association studies.*
Probabilistic Modeling in Genomics, Sep. 12–14, 2016, in Oxford, United Kingdom.

Grant activity

Current

Name	Funding agency	Period	Type	Role	Annual direct cost	Annual indirect cost
Doubly-robust variable selection in high dimensions	National Science Foundation	09/01/2023 - 08/31/2027	External	PI	\$46,154	\$28,846

Past

Name	Funding agency	Period	Type	Role	Annual direct cost	Annual indirect cost
Testing and estimation for multi-modality single cell genomics	National Science Foundation	08/01/2021-07/31/2024	External	PI	\$37,948	\$23,718
Statistical Software for Single Cell CRISPR Screens	Analytics at Wharton	2021-2023	Internal	PI	\$25,000	\$0